📄 PlacementConnect - Product Requirements Document (PRD)

1. Product Overview

PlacementConnect is a role-based campus recruitment application. It connects institute administrators, recruiting companies and students through a shared placement workflow. The product has a Vue frontend and a Flask backend. The frontend uses Vue Router, Pinia and Axios. The backend uses Flask-RESTX, Flask-SQLAlchemy and supporting Redis/Celery/mail/report components.

2. Product Goals

- Provide a single workspace for campus recruitment operations.
- Reduce manual coordination between placement office, companies and students.
- Enforce approvals and application eligibility consistently.
- Make recruitment status visible to relevant stakeholders.
- Support operational reporting, notifications and data exports.

3. Personas

Persona A — Placement Administrator

Needs to approve companies and drives, monitor students and companies, inspect applications, manage account access and review platform activity.

Persona B — Company Recruiter

Needs to maintain company information, publish job opportunities, review applicants, schedule interviews and record selection and offer outcomes.

Persona C — Student

Needs to maintain an accurate profile, discover relevant opportunities, understand eligibility and deadlines, apply or withdraw when permitted, and track progress.

4. Information Architecture / Modules

Module	Admin	Company	Student
Authentication	Login/logout	Register/login/logout	Register/login/logout

Dashboard	Platform and recruitment overview	Company drives and application metrics	Eligible opportunities and personal application progress
Profile	View/update/export	View/update/delete/export	View/update/delete/export
Company management	List, approve/reject/reapprove, disable/enable, blacklist/whitelist, delete	Own profile	Browse approved companies
Student management	List, disable/enable, blacklist/whitelist, delete	Applicant details in drive context	Own profile
Drive management	Review and approve/reject/reapprove; inspect	Create/update/delete/list own drives	Browse approved drives and details
Applications	Inspect application lists/details	Review and update own drive applicants	Apply, withdraw and track
Inbox	Notifications and attachments	Notifications and attachments	Notifications and attachments

5. Functional Requirements

5.1 Authentication and Access

- FR-001: The product shall allow Student and Company users to register.
- FR-002: The product shall allow users to log in using username or email, password and role-specific login flow.
- FR-003: The product shall reject login when the supplied role does not match the account role.
- FR-004: The product shall reject login for Disabled or Blacklisted accounts.

- FR-005: The product shall reject Company login while company approval is Pending or Rejected.
- FR-006: The product shall expose logout and current-user/session validation.
- FR-007: The frontend shall route authenticated users to role-appropriate dashboards and prevent cross-role route access.

5.2 Profiles

- FR-010: Admin shall be able to view and update the admin profile.
- FR-011: Company shall be able to view and update its company profile and use the profile deletion/export workflow.
- FR-012: Student shall be able to view and update personal, academic and resume/profile information, including education, experience and project records, and use deletion/export workflows.
- FR-013: The system shall validate profile data using server-side validation rules.

5.3 Admin Operations

- FR-020: Admin shall view a dashboard of placement activity.
- FR-021: Admin shall list companies and students.
- FR-022: Admin shall inspect drives and applications associated with a company or student.
- FR-023: Admin shall approve, reject or reapprove companies.
- FR-024: Admin shall enable, disable, blacklist, whitelist and delete company accounts.
- FR-025: Admin shall enable, disable, blacklist, whitelist and delete student accounts.
- FR-026: Admin shall approve, reject or reapprove placement drives.
- FR-027: Admin shall inspect application detail records.

5.4 Company Operations

- FR-030: Company shall view a dashboard of its drives, application and recruitment outcomes.
- FR-031: Company shall list its drives and view applicants for a drive.
- FR-032: Company shall create, update and delete its drives.
- FR-033: Company shall view application details for its own drives.
- FR-034: Company shall update application, interview and placement information according to allowed state transitions.
- FR-035: Company shall not access another company's drive or application data.

5.5 Student Operations

- FR-040: Student shall view a dashboard showing drive phases, eligibility and application-stage progress.

- FR-041: Student shall browse approved, active companies.
- FR-042: Student shall browse approved drives for an approved company.
- FR-043: Student shall view drive details and eligibility.
- FR-044: Student shall apply to a drive only if the application window is open, the student meets the minimum CGPA (if configured), and no application already exists.
- FR-045: Student shall withdraw an application only while the application window is open and the application status is Applied.
- FR-046: Student shall view application, interview and placement statuses.

5.6 Inbox, Notifications and Exports

- FR-050: Each role shall be able to view its inbox notification summaries and notification details.
- FR-051: Each role shall be able to retrieve permitted notification attachments.
- FR-052: The system shall expose notification status information.
- FR-053: The system shall support profile export and background report/export workflows where implemented by the relevant role services.
- FR-054: The system shall support background notification/reminder/report/export processing.

6. Non-Functional Requirements

ID	Requirement
NFR-01 Security	All protected API operations must enforce authentication and role/ownership authorisation server-side.
NFR-02 Data integrity	Enforce unique usernames/emails and one application per student-drive pair.
NFR-03 Validation	Validate inputs on the server; frontend validation is supplementary only.
NFR-04 Performance	Use dashboard caching where appropriate; invalidate cache when relevant records change.
NFR-05 Responsiveness	Core workflows should be usable on supported desktop and mobile

	browser sizes (verify through UI testing).
NFR-06 Reliability	Background tasks should expose failures and support retries/monitoring.
NFR-07 Time handling	Apply a consistent IST policy to application and interview date-times.
NFR-08 Maintainability	Keep role-specific API resources, schemas, services and frontend views modular.
NFR-09 Privacy	Student personal, academic and resume data must be disclosed only to authorised actors.
NFR-10 Accessibility	Forms, status indicators, navigation and error messages should be keyboard- and screen-reader-accessible.

7. Key User Journeys

Journey 1 — Company onboarding

Register → log in after approval → complete/update company profile → create drive → await admin approval → monitor drive and applicants.

Journey 2 — Student applies

Register → log in → complete profile → browse companies → browse drives → view eligibility and date window → apply → track status → withdraw if still Applied and within the application window.

Journey 3 — Recruiter processes an applicant

Open a drive → inspect applications → reject or shortlist → supply interview details when shortlisting → update interview outcome → if selected, supply joining date → update offer to Accepted or Declined.

Journey 4 — Admin governs the platform

Review company registrations → approve/reject → review submitted drives → approve/reject → monitor dashboard → manage user account status → inspect application details.

8. Product States and Status Vocabulary

- User account: Active, Disabled, Blacklisted.
- Company approval: Pending, Approved, Rejected.
- Drive approval: Pending, Approved, Rejected.
- Drive phase (derived): Upcoming, Ongoing, Closed.
- Application: Applied, Shortlisted, Rejected.
- Interview: Scheduled, Selected, Not Selected.
- Placement: Offered, Accepted, Declined.

These are separate state dimensions. For example, a company may have an Approved company profile while one of its drives is Pending approval.

9. UX Requirements

- Explain why a company cannot log in while approval is pending/rejected.
- Show eligibility and the reason a student cannot apply.
- Show application start and deadline with timezone.
- Confirm destructive actions (delete, withdraw, disable, blacklist).
- Display clear status labels and explain what each status means.
- Show empty, loading, success and error states on list and detail views.
- Explain that a shortlisted application requires interview details and selection requires a joining date.

10. Dependencies and Technical Context

- Frontend: Vue 3, Vite, Vue Router, Pinia, Axios.
- Backend: Flask, Flask-RESTX, Flask-SQLAlchemy.
- Supporting services: Redis, Celery, Flask-Mail/Mailpit integration, report/export generation and file storage.
- Data model: User, Company, Student, Education, Experience, Project, Drive, Application, Interview and Placement.

11. Product Gaps / Recommendations

- Add a visible profile-completion state and explicit required-field emphasis.
- Add audit history for approvals and recruitment status changes.
- Add status transition timeline to application details.
- Define whether drive edits after approval reset approval status.
- Add pagination, filtering and search requirements with performance targets.
- Add notification preferences and delivery/read tracking definitions.
- Add formal data retention, consent and privacy controls.